

Trustworthy Knowledge Graphs for Grid Digitalization: Standards-Native CIM/CGMES vs. LLM-Extracted Graphs in Power-System GraphRAG

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Abstract. As utilities adopt LLM-based assistants over CIM/CGMES models, a basic reliability question arises: should the underlying knowledge graph come from parsing the engineering standard, or from an LLM extraction over the same content? We answer this inside an otherwise frozen GraphRAG pipeline on two ENTSO-E CGMES conformity networks (MicroGrid, 233 objects; SmallGrid, 1,398 objects), using the CGMES export itself as a schema-grounded oracle for fidelity. LLM extraction is precise but incomplete (near-zero schema contradiction, 34–53% of the standard’s structure recovered); raw CIM is exact but retrieval-verbose; on the larger network the LLM-extracted graph wins under a fixed 8,000-character context budget (98% for the LLM-extracted graph vs. 48% for the standards-native graph on SmallGrid). The advantage is representational selectivity, not higher graph correctness. A deterministic semantic compression of CGMES recovers part of that advantage at zero LLM cost. Code, prompts, the schema-grounded scorer, and audit trails are released as a reusable benchmark for trustworthy grid-AI evaluation.

Index Terms. Smart grid AI, CIM/CGMES, IEC 61970, knowledge graph, GraphRAG, power-system data standards, trustworthy AI, grid digitalization, LLM reliability, semantic compression.

I. INTRODUCTION

Grid digitalization is driving rapid adoption of LLM-based assistants over utility data, but the reliability of these assistants depends critically on the knowledge graph they retrieve from. In the power-system setting that graph can come from two very different sources: the deployed engineering standard itself (CIM/CGMES, IEC 61970), or an LLM extraction over the same content. This paper measures the difference, in a frozen GraphRAG pipeline, with the standard serving as schema-grounded ground truth, so that grid operators have a controlled basis for choosing between the two when deploying trustworthy AI on top of CIM/CGMES models.

Graph-based retrieval-augmented generation (GraphRAG) augments an LLM reader with a knowledge graph over the target corpus, and its strongest reported gains cluster in relation-dense engineering domains [1–4]. In essentially all power-domain GraphRAG and RAG work, including the closest in-domain prior art HG-RAG [5] and the vector-RAG grid-code

system GridCodex [6], any graph used is constructed by an LLM from text, and no engineering standard is used as ground truth or alternative construction path; the utility context for these assistants is surveyed in [7].

Power systems are unusual in that a curated, machine-readable relational model already exists as a deployed engineering standard: IEC 61970 defines the Common Information Model (CIM) [8]; CGMES is its exchange profile; ENTSO-E publishes conformity test models [9]. Parsing CGMES into a graph is not new: CIM-as-ontology is established practice [10–12]. The novelty here is using the same CGMES export simultaneously as (a) a *construction source* for one graph arm and (b) a *schema-grounded oracle* for judging an LLM-extracted graph arm, inside one otherwise frozen GraphRAG pipeline. Compared to general-domain KG-construction studies that find LLM extraction strong on relevance but weak on completeness [13], [14], our contribution is a power-domain, construction-source-controlled paired contrast with a real standard as oracle and build cost as a first-class measured axis.

Research questions. **RQ1 (fidelity):** how much of what the LLM extracts is wrong, and how much of the schema does it miss? **RQ2 (downstream):** does graph construction source change paired QA accuracy in an otherwise frozen pipeline? **RQ3 (cost):** what is the exact token gap between the two construction paths?

II. SETUP AND METHOD

A. Two Graphs, One Frozen Pipeline

For each test network we build two graphs (Fig. 1). G_{STD} (**standards-native**) is parsed deterministically from the CGMES EQ/TP/SSH RDF/XML; nodes are CIM objects (mRID, class, name, numeric attributes), edges are the export’s own association predicates (Terminal.ConductingEquipment, Equipment.EquipmentContainer, PowerTransformerEnd.PowerTransformer, etc.). MicroGrid: 233 nodes, 362 edges; SmallGrid: 1,398 nodes,

2,184 edges. Zero LLM involvement. G_{LLM} (**LLM-extracted**) is built by LightRAG 1.5.1 (LightRAG defaults: two gleaning passes, 512-token chunks, 64-token overlap) with an Azure gpt-5.1 extraction model, over a deterministic, template-based natural-language rendering of the same CGMES model (MicroGrid: 2,299 words; SmallGrid: 14,224 words). The corpus is generated without any LLM, removing the circularity of having the extraction model author its own input; the clean corpus deliberately favors extraction, so measured error rates are a plausible lower bound.

B. Edge Taxonomy and Scoring (RQ1)

Each extracted entity name is aligned to CGMES objects by exact normalized-name match, class-qualifier affix stripping, then unique fuzzy match (Levenshtein ≤ 2). From the raw CGMES associations we derive five support relations by fixed rules (ELECTRICALLYCONNECTED, CONNECTSTONODE, CONTAINEDIN, HASPART, ATVOLTAGE). Each extracted edge with both endpoints aligned is classified MATCH (any candidate pair supported), Type B (unsupported but corpus-expressed; excluded), Type A (unsupported, not corpus-expressed, label maps to a schema relation: asserts structure the model does not contain), or Type C (no warrant). The error rate is $(A+C)/(M+A+C)$. We also report *topological recall*: the fraction of supported CGMES pairs, over entities the LLM extracted at all, that appear as some extracted edge.

C. Paired QA (RQ2)

50 items per network are generated deterministically from CGMES: 25 single-hop (S1) attribute/association lookups and 25 multi-hop (S2: containment chains, line-endpoint resolution, counts, region tracing). One code path serves both arms: a symmetric 2-hop retriever with an identical 8,000-character context cap and degree-based fallback. Reader: local Qwen2.5-7B-Instruct [15] in bfloat16, greedy decoding, identical prompts, on one NVIDIA GB10 (128 GB unified CPU-GPU memory). Primary scoring is deterministic against the gold spec; a secondary cross-family judge pair (Azure gpt-5.1 and Claude Haiku) agreed with each other on 200/200 verdicts and with the deterministic scorer on 199/200. We use McNemar’s exact test on paired binary verdicts, Holm correction across the three registered contrasts (All, S1, S2), Wald 95% CIs on the paired-difference, and pre-registered minimum detectable effects (23 pp at $n=50$; 32 pp per 25-item stratum, $\alpha=0.05$, power 0.80): any sub-MDE effect is reported as inconclusive regardless of nominal p . A registered human spot-check ($n=10$ items, 20 paired verdicts) agreed with the deterministic scorer on 20/20; per-item annotations are included in the released artifact bundle. A first run was retracted after an internal audit found protocol violations; the corrected run reported here, the retracted artifacts, and a full deviations register are released. The full artifact bundle (CGMES→graph builders, LightRAG extraction harness, deterministic corpus and QA generators,

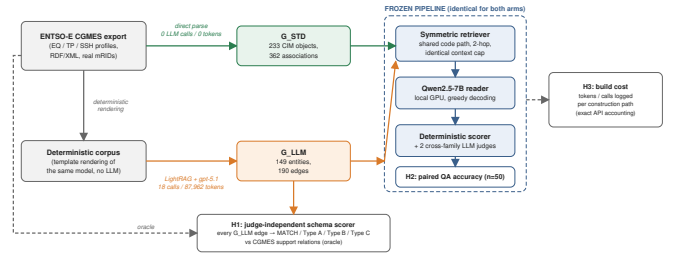


Fig. 1. Two graph-construction paths, one frozen GraphRAG pipeline. G_{STD} is parsed deterministically from the CGMES export (0 LLM calls); G_{LLM} is built by LightRAG from a deterministic textual rendering of the same model. Both feed an identical symmetric retriever and the same Qwen2.5-7B reader; only the graph differs. A judge-independent scorer additionally classifies every G_{LLM} edge against the CGMES schema.

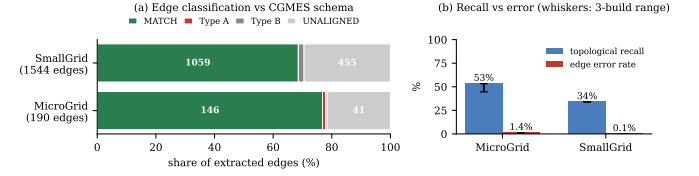


Fig. 2. RQ1 schema fidelity. (a) Almost all aligned edges are schema-supported. (b) The failure mode is incompleteness, not hallucination: topological recall vs edge error rate (whiskers: range over independent LightRAG builds).

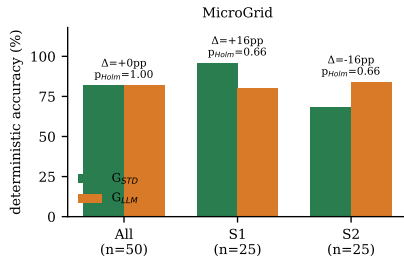
schema-grounded edge scorer, reader/judge prompts, per-item audit trails, pre-registration, deviations register, and human spot-check annotations) is released at <https://github.com/jmanoharan/cgmeg-graphrag-bench> and permanently archived at <https://doi.org/10.5281/zenodo.20671115>.

III. RESULTS

Two axes. We separate *graph fidelity* (does the graph faithfully encode CGMES?) from *retrieval efficiency* (how much useful signal fits inside a fixed context budget?). G_{STD} has higher fidelity by construction; G_{LLM} has higher retrieval efficiency under the frozen budget. On the larger network these two axes disagree.

A. RQ1: Precise but Incomplete

On MicroGrid, LightRAG extracted 149 entities and 190 edges; 80.5% of entities aligned. Of the 148 scored edges, 146 are MATCH, 2 Type A, 1 Type B, 0 Type C: edge error **1.4%** (2/148); label fidelity 100%; topological recall **53.4%** (159/298); 19.5% of extracted entities are unalignable (mangled strings such as “Usbar_6”, free-floating literals such as “380 KV”). Across three independent LightRAG builds, error 0.0 to 1.4%, recall 44.6 to 53.4%, alignment 67.5 to 80.5%. SmallGrid amplifies the pattern: 80.8% entity alignment, error **0.09%** (1/1,060; 29 Type B excluded), label fidelity 98.9%, but topological recall falls to **34.0%** (929/2,736). Precision is stable; *completeness degrades*. The pre-registered hypothesis (“a non-trivial fraction of extracted edges is schema-contradictory”) is refuted as worded: hallucinated structure is not the failure mode, missing structure is.



All stratum deltas are below the pre-registered MDE (23pp full set / 32pp per stratum) — reported as inconclusive.

Fig. 3. RQ2 downstream QA accuracy by stratum on MicroGrid (deterministic scoring). On MicroGrid, no contrast reaches the pre-registered MDE; the SmallGrid result is tabulated in Table I.

TABLE I

H2 DOWNSTREAM QA: PAIRED DETERMINISTIC ACCURACY PER CASE AND STRATUM. $\Delta = G_{STD} - G_{LLM}$; CI = WALD 95% INTERVAL ON THE PAIRED DIFFERENCE; b/c = DISCORDANT PAIRS (G_{STD} -ONLY / G_{LLM} -ONLY CORRECT). PRE-REGISTERED MDES: 23 PP AT $n=50$ (ALL), 32 PP PER 25-ITEM STRATUM. ON MICROGRID ALL $|\Delta|$ ARE BELOW THE MDE AND ARE REPORTED AS INCONCLUSIVE; ON SMALLGRID THE ALL AND S2 CONTRASTS EXCEED THE MDE AND ARE REPORTED AS SIGNIFICANT AFTER HOLM CORRECTION (BOLD p_{Holm}).

Case	Stratum	n	G_{STD}	G_{LLM}	Δpp	CI _{95pp}	b/c	p	p_{Holm}
MicroGrid	All	50	82	82	+0.0	[-13.6, +13.6]	6/6	1.000	1.000
MicroGrid	S1	25	96	80	+16.0	[-2.2, +34.2]	5/1	0.219	0.656
MicroGrid	S2	25	68	84	-16.0	[-34.2, +2.2]	1/5	0.219	0.656
SmallGrid	All	50	48	98	-50.0	[-64.9, -35.1]	1/26	$< 10^{-4}$	$< 10^{-4}$
SmallGrid	S1	25	84	96	-12.0	[-28.9, +4.9]	1/4	0.375	0.375
SmallGrid	S2	25	12	100	-88.0	[-100.7, -75.3]	0/22	$< 10^{-4}$	$< 10^{-4}$

B. RQ2: Downstream Accuracy Is Scale-Dependent

Table I and Fig. 3 report paired accuracy.

MicroGrid: no detectable difference. Both arms score **82%** overall ($n=50$; McNemar exact $p=1.0$; CI $[-13.6, +13.6]$ pp). The design detects ≥ 23 pp; this is “no difference of at least 23 pp detected,” not equivalence. Strata are opposite-signed and both below the 32 pp MDE (S1: 96% vs. 80%, +16 pp; S2: 68% vs. 84%, -16 pp; Holm $p=0.66$).

SmallGrid: the LLM-extracted graph wins under the fixed 8,000-character context budget; the advantage is representational selectivity and compression, not higher graph correctness (G_{STD} contains strictly more correct structure; §III-A). G_{LLM} reaches **98%** overall versus **48%** for G_{STD} (-50 pp; CI $[-64.9, -35.1]$ pp; McNemar $p < 10^{-4}$), driven by the multi-hop stratum (S2: 100% vs. 12%, -88 pp, exceeding the registered MDE; S1: 96% vs. 84%, n.s.). Diagnosis: **45 of 50** G_{STD} contexts hit the cap (median 8,020 chars) versus 6 of 50 for G_{LLM} (median 2,400). The raw CIM representation routes every connection through Terminal objects, so its 2-hop neighborhoods on a 1,398-node graph overflow the budget before the relevant relations are rendered.

Secondary baseline: compressed standards-native graph (G_{HYB}). A post-hoc, non-registered third arm, the *compressed standards-native graph* G_{HYB} , keeps the CGMES objects of G_{STD} but replaces raw CIM association edges with our five derived support relations (zero LLM tokens). G_{HYB} ties the *best single-hop attribute accuracy* (96% on SmallGrid

TABLE II

H3 GRAPH CONSTRUCTION COST (EXACT API USAGE ACCOUNTING). WALL-CLOCK FOR G_{LLM} IS DOMINATED BY CLOUD API LATENCY AND IS NOT A HEADLINE METRIC; THE STRUCTURAL ASYMMETRY IS THE TOKEN COLUMN.

Case	Build	Wall	LLM calls	Tokens	Nodes	Edges
MicroGrid	G_{STD} (CGMES parse)	10 ms	0	0	233	362
MicroGrid	G_{LLM} (LightRAG)	78 s	18	87,962	149	190
SmallGrid	G_{STD} (CGMES parse)	77 ms	0	0	1398	2184
SmallGrid	G_{LLM} (LightRAG)	644 s	112	594,617	1179	1544

S1, equal to G_{LLM}), but S2 collapses to 12% on SmallGrid because shared-node closure produces dense busbar cliques that re-overflow the cap (46/50 truncated; overall SmallGrid 54%). MicroGrid: S1 84%, S2 64%, overall 74%. Compression is necessary but not sufficient; *selectivity* is the missing ingredient.

Table III summarizes the three-arm trade-off.

C. RQ3: Cost Asymmetry

Table II reports exact build costs. G_{STD} is parsed in 10 ms (MicroGrid) and 77 ms (SmallGrid) with zero LLM calls. G_{LLM} cost 18 calls / 87,962 tokens on the 2,299-word MicroGrid corpus and 112 calls / 594,617 tokens on the 14,224-word SmallGrid corpus (corpus grows 6.2 $\times$; tokens grow 6.8 $\times$).

D. Error Analysis

G_{LLM} fails by *numeric attribute migration*: the extracted graph answers “length of NL-Line_4?” with 23 km, the length of NL-Line_3 (gold: 22 km), and similarly migrates resistance and reactance values between BE-Line_3, BE-Line_4, and BE-Line_7. **This is the operational failure mode practitioners must guard against.** G_{STD} fails through *representational indirection and budget overflow*. A 3-hop sensitivity study on MicroGrid S2 confirms the mechanism: G_{STD} rises 68% to 76%, G_{LLM} falls 84% to 76% as larger neighborhoods dilute its context.

IV. DISCUSSION AND CONCLUSION

Parse the standard, but do not retrieve over it raw.

For CGMES-covered content, keep CGMES as the authoritative source and *compress it deterministically for retrieval*; our derived support-relation layer is a first step. The realistic use case for LLM extraction is the content CGMES does not cover (maintenance records, incident reports, grid codes, vendor manuals); for those, our results suggest the failure profile to engineer against is omission and numeric attribute migration, not invented structure.

Threats to validity. (i) *External validity*: two conformity test networks, one corpus style, one reader; we claim nothing about TSO-scale models. (ii) *Corpus realism*: the deterministic corpus favors extraction; RQ1 numbers are plausible lower bounds for real documentation. (iii) *Statistical power*: $n=50$ detects only ≥ 23 pp overall; stratum observations are inconclusive unless flagged otherwise. (iv) *Retrieval design*: the symmetric 2-hop retriever is the clean instrument for the

TABLE III

THREE-ARM CONTRAST. BUILD TOKENS ARE EXACT API ACCOUNTING (MICROGRID / SMALLGRID) FROM TABLE II. G_{HYB} (COMPRESSED STANDARDS-NATIVE GRAPH) IS A POST-HOC, NON-REGISTERED BASELINE THAT ADDS DETERMINISTIC SHORTCUT EDGES TO G_{STD} AT ZERO LLM COST.

Arm	Build source	Tokens (MG / SG)	Strength	Weakness
G_{STD}	CGMES RDF/XML parse	0 / 0	Highest fidelity; full CIM coverage; every edge traceable to an mRID	Retrieval-verbose; 45/50 SmallGrid contexts truncated at the 8,000-char cap
G_{LLM}	LightRAG over CGMES prose	87,962 / 594,617	Selective rendering; fits the cap; best overall SmallGrid QA (98%)	Topological recall 53.4% (MG) / 34.0% (SG); numeric attribute migration between similarly named lines
G_{HYB}	CGMES parse plus derived shortcut edges	0 / 0	Tied-best SmallGrid single-hop attribute accuracy (96%, equal to G_{LLM}); zero LLM cost	Multi-hop collapses (SmallGrid S2 = 12%); shared-node closure re-overflows the cap

graph-source effect but handicaps LightRAG’s native dual-level retrieval; depth and native-retrieval sensitivity studies are future work. (v) *Builder-as-judge*: the Azure gpt-5.1 judge is from the G_{LLM} builder’s family, but both judges agreed on 200/200 verdicts and with the deterministic scorer on 199/200. (vi) *Stochastic replication*: three independent MicroGrid builds, single SmallGrid build; the directional findings are mechanism-grounded. (vii) *Retracted prior run*: disclosed in the released register; supplementary `DEVIATIONS.md` documents the audit trail.

Conclusion. A construction-source-controlled paired contrast shows that LLM extraction is precise but incomplete and that the standards-native graph, while higher-fidelity, suffers under a fixed retrieval budget. Selective deterministic rendering of CGMES is the natural next step, and it costs nothing in LLM tokens.

REFERENCES

- [1] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, D. Metropolitansky, R. O. Ness, and J. Larson, “From local to global: A graph rag approach to query-focused summarization,” 2024, arXiv:2404.16130.
- [2] Z. Guo, L. Xia, Y. Yu, T. Ao, and C. Huang, “Lightrag: Simple and fast retrieval-augmented generation,” 2024, arXiv:2410.05779.
- [3] S. Ahmad, Z. Nezami, M. Hafeez, and S. A. R. Zaidi, “Benchmarking vector, graph and hybrid retrieval augmented generation (rag) pipelines for open radio access networks (oran),” 2025, arXiv:2507.03608.
- [4] S. Knollmeyer, O. Caymazer, and D. Grossmann, “Document graphrag: Knowledge graph enhanced retrieval augmented generation for document question answering within the manufacturing domain,” *Electronics (MDPI)*, 2025.
- [5] Z. Shen, X. Cai, B. Ni, Z. Meng, Z. Huang, and T. Yu, “Hg-rag: Hierarchical graph-enhanced retrieval-augmented generation for power systems,” *Electronics (MDPI)*, 2026.
- [6] J. Shi, Y. Cheng, F. Zhang, M. Jiang, J. Lin, and Y. Shen, “Gridcodex: A rag-driven ai framework for power grid code reasoning and compliance,” 2025, arXiv:2508.12682.
- [7] EPRI, “Benchmarking large language models for the electric power sector,” Electric Power Research Institute, Tech. Rep., 2025, ePRI Product ID 3002034347.
- [8] International Electrotechnical Commission, “IEC 61970: Energy management system application program interface (EMS-API) - part 301: Common information model (CIM) base,” IEC Standard, 2020.
- [9] ENTSO-E, “Common grid model exchange standard (CGMES) v2.4.15 - conformity assessment scheme test configurations,” European Network of Transmission System Operators for Electricity, Tech. Rep., 2014, microGrid/MiniGrid/SmallGrid test models; model files state ENTSO-E ownership and attribution requirements. Mirrored in the powstyle-core repository (cgmes-conformity resources). [Online]. Available: <https://www.entsoe.eu/digital/common-information-model/>
- [10] M. Gaha, A. Zinflou, C. Langheit, A. Bouffard, M. Viau, and L. Vouligny, “An ontology-based reasoning approach for electric power utilities,” in *Web Reasoning and Rule Systems (RR 2013)*, ser. Lecture Notes in Computer Science, vol. 7994. Springer, 2013.
- [11] A. Schumilin, K.-U. Stucky, F. Sinn, and V. Hagenmeyer, “Towards ontology-based network model management and data integration for smart grids,” in *2017 Workshop on Modeling and Simulation of Cyber-Physical Energy Systems (MSCPES)*. IEEE, 2017.
- [12] International Electrotechnical Commission, “IEC 61970-501: Energy management system application program interface (EMS-API) - part 501: Common information model resource description framework (CIM RDF) schema,” IEC Standard, 2006.
- [13] R. Chen, W. Jiang, C. Qin, B. Xiong, F. Liausvia, D. Choi, and B. K. Quek, “Are large language models effective knowledge graph constructors?” 2025, arXiv:2510.11297.
- [14] N. Mihindukulasooriya, S. Tiwari, C. F. Enguix, and K. Lata, “Text2KGBench: A benchmark for ontology-driven knowledge graph generation from text,” in *Proceedings of the 22nd International Semantic Web Conference (ISWC)*, 2023, arXiv:2308.02357.
- [15] Qwen Team, “Qwen2.5 technical report,” 2024. [Online]. Available: <https://arxiv.org/abs/2412.15115>